

PRODUCT BACKLOG

Project: Dental Fable

Management System for a Pediatric Dental Clinic

Team: CodeXplorers | Version: 2.0 | Date: May 6th 2026
Technologies: PHP · MongoDB · Bootstrap 5 · JavaScript · Docker

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1. Project Description

Fábula Dental is a web-based clinical management system specializing in pediatric dentistry. The platform allows users to manage patients, track payments, and manage dental supply inventory, with differentiated access based on user role (Administrator, Dentist, Receptionist).

Technology Stack:

- Backend: PHP 8.x with MVC architecture
- Database: MongoDB Atlas (collections : patients, payments, supplies)
- Frontend: Bootstrap 5.3, Vanilla JavaScript, HTML5, CSS3
- Infrastructure : Docker (Dockerfile) including)
- Authentication: PHP sessions with role control

System Actors :

- Administrator: Manages the inventory of dental supplies
- Dentist: Registers and manages patient information
- Receptionist: Registers and manages patient payments
- Visitor: Consult the public information on treatments

2. User Stories

User stories describe the functionalities of the system from the perspective of each actor, following the standard agile format: As [role], I want [action], for [benefit].

ID	As...	Wanna ...	For...	Acceptance Criteria
US-01	System user	Log in with username and password	access the module corresponding to my role	• Redirection based on role after successful login • Error message for invalid credentials • Fields cannot be empty
US-02	Dentist	Register a new patient with ID number, name, date of birth, and reason for consultation	have the patient's complete medical history in the system	• ID card with exactly 10 digits mandatory • Mandatory legal representative if patient < 18 years • Successful registration saves to MongoDB and redirects to success page
US-03	Dentist	View, edit, and delete patients from the list	keep patient information up-to-date and organized	• Table loading patients from the API (GET) • Online editing sends PUT to the API • Deletion sends DELETE and removes the row without reloading
US-04	Receptionist	record a payment indicating patient, amount, type and payment method	to keep track of the clinic's income and payments	• Payment type: Deposit or Final • If type = Final, status = Completed; otherwise, status = Partial • Amount must be greater than 0 and date cannot be empty

ID	As...	Wanna ...	For...	Acceptance Criteria
US-05	Receptionist	View the complete payment history with its current status	to have financial traceability and know which patients have outstanding payments	- Table shows all payments (GET to API) - Status with differentiated color: Completed (green), Partial (yellow), Pending (red) - Records can be edited and deleted
US-06	Administrator	Record dental supplies with name, quantity, cost, and dates	control the clinic's inventory and avoid running out of supplies	- Amount minimum : 1 unit - Expiration date $\geq$ order date - Automatically calculated status: Current / Expiry / Expired
US-07	Administrator	View inventory with alerts for products nearing their expiration date	anticipate the need to replenish materials before they run out or expire	- Status 'To Expire' if ≤ 30 days remaining - Status 'Expired' if the date has already passed - Differentiated visual indicator in the table
US-08	Clinic visitor	View the list of available treatments without logging in	I would like to know about the services before making an appointment.	- Accessible from the login header - Displays a table with the name, description, and post-treatment recommendations - It does not require authentication

3. Product Backlog

The Product Backlog below is ordered by epics and priority.
Story points follow the Fibonacci scale (1, 2, 3, 5, 8).

High Priority **Medium Priority** **Low Priority** **Filled** **Earring**

ID	User Story	Description / Acceptance Criteria	Acceptance Criteria	Priority	Points	State
PB-01	Role-based login	<i>As a system user, I want to log in with my credentials to access the features of my specific role.</i>	The system validates username/password. It redirects to the correct view based on the user's role (Admin , Dentist, Receptionist). It displays an error if the credentials are incorrect .	High	5 pts	Filled
PB-02	Role-based access control	<i>As the system administrator, I want each user to only have access to the modules that correspond to them in order to protect the information.</i>	The Administrator only has access to Inventory. The Dentist only has access to Patients. The Receptionist only has access to Payments.	High	3 pts	Filled
PB-03	Log out	<i>As an authenticated user, I want to securely log out to protect the clinic's information.</i>	The 'Log Out' button destroys the PHP session and redirects to the login .	High	2 pts	Filled

ID	User Story	Description / Acceptance Criteria	Acceptance Criteria	Priority	Points	State
PB-04	New patient registration	<i>As a dentist, I want to register a new patient with their personal details so that I have their information in the system.</i>	The form captures: ID number (10 digits), full name, date of birth, phone number, reason for consultation, and legal representative (required if under 18 years of age). The system It is stored in MongoDB.	High	8 pts	Filled
PB-05	List of patients	<i>As a dentist, I want to see the list of all registered patients in order to search for and manage their data.</i>	The table displays all patients. Each record can be edited or deleted. The data is loaded. from the API.	High	5 pts	Filled
PB-06	Patient data editing	<i>As a dentist, I want to edit the data of an existing patient to keep their information up to date.</i>	Clicking "Edit" makes the fields editable. The "Save" button sends a PUT request to the API. Changes are reflected in the list in real time.	High	5 pts	Filled
PB-07	Patient removal	<i>As a dentist, I want to delete a patient record from the system when necessary.</i>	The delete button sends a DELETE request to the API. The record disappears from the list without reloading the page.	Average	3 pts	Filled
PB-08	Validation of legal representative for minors	<i>As a dentist, I want the system to require me to enter a legal representative if the patient is</i>	The system calculates age based on date of birth. If the person is under 18 and has no legal guardian, registration will fail with an error message.	High	3 pts	Filled

ID	User Story	Description / Acceptance Criteria	Acceptance Criteria	Priority	Points	State
		<i>under 18 years of age, in order to comply with legal requirements.</i>				
PB-09	Payment record	<i>As a receptionist, I want to record a patient's payment, indicating the type, method, and amount, to keep financial control.</i>	The form captures: Patient ID number, date, amount, payment type (Deposit/Final) and method (Cash/Transfer/Card). If the payment is 'Final', the status is marked as 'Completed'.	High	8 pts	Filled
PB-10	Payment history	<i>As a receptionist, I want to see the history of all recorded payments to monitor the clinic's income.</i>	shows all payments with their status (Pending, Partial, Completed) . edit and delete records .	High	5 pts	Filled
PB-11	Paid edition existing	<i>As a receptionist, I want to edit a recorded payment to correct errors or update the status.</i>	The fields are editable online. The system sends PUT requests to the API . Changes are reflected immediately .	Average	3 pts	Filled
PB-12	Visual indicator of payment status	<i>As a receptionist, I want to see the payment status</i>	Pending = red/orange indicator. Partial = yellow indicator. Completed = indicator green .	Average	2 pts	Earring

ID	User Story	Description / Acceptance Criteria	Acceptance Criteria	Priority	Points	State
		<i>with distinctive colors to quickly identify pending or completed payments.</i>				
PB-13	Supply record	<i>As an administrator, I want to register a new dental supply with its name, quantity, cost, and dates to manage the inventory.</i>	The form captures: name, quantity (min. 1), unit cost, order date, and expiration date. The expiration date must be later than the order date.	High	8 pts	Filled
PB-14	Inventory list with status	<i>As an administrator, I want to see the inventory of supplies with their expiration status in order to make replenishment decisions.</i>	Displays: name, quantity, cost, dates. Automatic status: 'Valid', 'Due' (≤30 days) or 'Expired'.	High	5 pts	Filled
PB-15	Product Expiration Alert	<i>As an administrator, I want the system to automatically identify supplies that are about to expire so they can be replenished on time.</i>	The system calculates the difference between the current date and the expiration date. If it is ≤ 30 days, the status is 'Due' with a visual alert indicator.	High	3 pts	Filled

ID	User Story	Description / Acceptance Criteria	Acceptance Criteria	Priority	Points	State
PB-16	Editing and deleting supplies	<i>As an administrator, I want to edit or delete supplies from the inventory to keep the information up to date.</i>	Edit online using PUT to the API. Delete using DELETE. The list is updated without reloading.	Average	3 pts	Filled
PB-17	Treatments page	<i>As a visitor to the clinic, I want to see the available treatments without having to log in to learn about the services.</i>	The treatments page is accessible from the login page . It displays a table with: name, description, and post-treatment recommendations .	Average	3 pts	Filled
PB-18	Patient search and filtering	<i>As a dentist, I want to search for patients by name or ID number to quickly find their information in a large list.</i>	Real-time search field that filters the table. Works by name and ID number.	Average	5 pts	Earring
PB-19	Dashboard with statistics	<i>As an administrator, I want to see summarized metrics on the main dashboard (total patients, monthly revenue,</i>	Cards with dynamic counters. Data obtained from the API in real time.	Low	8 pts	Earring

ID	User Story	Description / Acceptance Criteria	Acceptance Criteria	Priority	Points	State
		<i>supplies due to expire) for a quick overview of the clinic's status.</i>				
PB-20	Secure database authentication	<i>As the system administrator, I want user credentials to be stored in the database with encrypted passwords for added security.</i>	Credentials are stored in MongoDB using bcrypt hashing . The login queries the database instead of hardcoded values .	High	8 pts	Earring
PB-21	Exporting reports	<i>As a receptionist or administrator, I want to export payment and inventory lists in PDF or Excel format for reporting purposes.</i>	Export button in the lists. Generates a downloadable file with the table data.	Low	8 pts	Earring

4. Sprint Planning

The following table summarizes the distribution of Backlog items in development Sprints , with the objective of each iteration and the current progress status.

Sprint	Items Included	Sprint Objective	Points	State
Sprint 1	PB-01, PB-02, PB-03, PB-04, PB-08, PB-17	Functional authentication, patient registration with validations, and a public treatment page.	19	Filled
Sprint 2	PB-05, PB-06, PB-07, PB-09, PB-10, PB-13, PB-14, PB-15	Complete CRUD for patients, payment registration, inventory management with expiration alerts.	37	Filled
Sprint 3	PB-11, PB-12, PB-16	Payment and inventory refinement: editing, deletion, and visual status indicators.	8	Filled
Sprint 4	PB-18, PB-19, PB-20	Security improvements (credentials in the database), patient search, and statistical dashboard .	21	Planned
Sprint 5	PB-21	Export of reports in PDF/Excel for payments and inventory.	8	Planned

Total story points: 93 points | **Completed (Sprints 1–3):** 64 points | **Pending (Sprints 4–5):** 29 points

5. Technical and Non-Functional Requirements

5.1 Implemented Functional Requirements

- RF-01: Authentication system with differentiated roles (Admin , Dentist, Receptionist)
- RF-02: Complete CRUD of patients with Ecuadorian ID validation (10 digits)
- RF-03: Validation of legal representative for patients under 18 years of age
- RF-04: Complete CRUD of payments with types (Deposit/Final) and methods (Cash/Transfer/Card)
- RF-05: Automatic update of payment status (Pending → Partial → Completed)
- RF-06: Complete CRUD of supplies with automatic expiration status calculation
- RF-07: Automatic alerts for products with an expiration date ≤ 30 days
- RF-08: Public treatment page accessible without authentication
- RF-09: RESTful APIs for each module (GET, POST, PUT, DELETE)

5.2 Non- Functional Requirements

- RNF-01 Security: PHP session control by role. Views are inaccessible without an active session.
- RNF-02 Usability: Responsive interface with Bootstrap 5. Compatible with mobile and desktop devices
- RNF-03 Performance: CRUD operations do not require a full page reload (fetch API)
- RNF-04 Scalability: NoSQL database (MongoDB) allows horizontal scaling
- RNF-05 Portability: Containerized with Docker for deployment in any environment
- RNF-06 Maintainability: MVC architecture with clear separation of models, views, and controllers
- RNF-07 Validation: Validation in both frontend (JavaScript) and backend (PHP)

6. Definition of Done (DoD)

A backlog item is considered DONE when it meets all of the following criteria:

- The code is implemented and working in the development environment
- The acceptance criteria have been verified manually.
- The functionality has been tested with real data in MongoDB Atlas
- There are no console errors or unhandled exceptions.
- The code follows the project's MVC structure (models /, controllers /, views /)

- The interface is responsive and visually consistent with the design of Dental Fable
- The functionality has been reviewed and approved by the CodeXplorers team